UAV LAB
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Vision-Language Model Fallback for Airborne Object Classification

Overcoming Closed-Vocabulary Failure in Deep Learning Detectors

Project Documentation

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Project Repository

https://github.com/SYED-IJTABA-RIZVI/UAV-Dependability-lab-Project

Datasets

RGB Dataset:
https://kaggle.com/datasets/2dcbe85ec4b4fa99410f6a4ecec9a2ff2186072b5a646c101495c4005eb54cb6

Infrared (Thermal) Dataset:
https://www.kaggle.com/datasets/34648aae4c599a4a6ab7e210ea19dd76b438ca95889e219b2f96cdce6136af69

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Abstract

Reliable detection and classification of low-altitude aerial objects remains a challenging problem, owing to small target size, visual similarity between object categories, environmental variability, and differences across sensing modalities. Conventional closed-vocabulary object detectors perform well on classes represented during training but exhibit reduced generalization when confronted with unseen object appearances or conditions outside the training distribution.

This project presents a comparative investigation of deep-learning-based and vision-language approaches to aerial-object detection and classification using RGB and thermal imagery. A unified dataset was constructed from multiple publicly available sources and standardized around four target classes: airplane, bird, drone, and helicopter. Five supervised object detectors — YOLOv8, YOLOv10, YOLO11, YOLO12, and RF-DETR — were trained and evaluated under a common experimental protocol. In parallel, four pretrained Vision-Language Models (VLMs) — BLIP-2, Qwen, DeepSeek, and InternVL — were assessed in a zero-shot setting on the same classification task.

Model performance is evaluated using a comprehensive set of metrics, including precision, recall, F1-score, Jaccard index, mean Average Precision, per-class performance, and invalid-output rate. Additional analysis examines prediction distributions, model behaviour, and failure cases to characterize the reliability and limitations of both supervised detectors and zero-shot VLMs. RGB and thermal performance are analysed independently to quantify the effect of sensing modality on detection and classification outcomes.

The resulting comparative framework offers a systematic assessment of modern object detectors and Vision-Language Models for low-altitude aerial-object recognition, and establishes the empirical foundation for a future hybrid system in which a supervised detector is supplemented by a VLM-based fallback mechanism under uncertain or ambiguous conditions.

Keywords: Aerial object detection, counter-UAV, YOLO, RF-DETR, vision-language models, zero-shot classification, RGB imaging, thermal imaging, multimodal learning.

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RCS Radar Cross-Section.

1. Introduction

1.1 Background and Motivation

The rapid growth of UAVs has enabled applications in photography, inspection, agriculture, logistics, and surveillance, but their low cost and autonomy have also introduced new security risks for airports, military sites, and urban areas.

Detecting low-altitude aerial objects is difficult: UAVs appear as small, distant targets with high variation in shape, scale, and orientation, and are easily confused with birds, helicopters, and airplanes. Radar handles long-range detection but struggles with small UAVs due to clutter and low radar cross-section (Radar Cross-Section (RCS)). Vision-based systems add complementary appearance information but degrade under adverse conditions, as summarized below.

Condition	**Effect on RGB Detection**
Nighttime	Low contrast, weak feature visibility
Fog	Reduced visibility and range
Rain	Lens contamination and blur
Smoke/haze	Indistinct silhouettes
Low illumination	Poor signal-to-noise ratio
Small/distant targets	Insufficient pixel detail

Table 1.1: Environmental and target-related factors affecting RGB-based aerial object detection

Thermal imaging complements RGB by sensing emitted infrared radiation, remaining usable in darkness at the cost of lower resolution and texture detail, motivating a multimodal RGB-thermal evaluation.

Deep learning has driven major progress in detection, with one-stage detectors such as YOLO enabling efficient inference and transformer-based detectors providing an alternative architecture. However, these supervised detectors operate with a fixed vocabulary defined by their training data. Vision-Language Models (VLMs) provide an alternative through visual and language understanding, enabling zero-shot classification without task-specific training.

This project therefore investigates a comparative multimodal aerial-object detection framework, evaluating multiple detector families and VLMs under a shared experimental setup rather than focusing on a single architecture, culminating in a deployable tool that integrates both model families (Chapter 6, Section 6.7).

1.2 Problem Statement

Detector performance can degrade for small, distant, occluded, poorly lit, or previously unseen targets, while closed-vocabulary detectors are restricted to the classes represented during training. VLMs provide a potential zero-shot alternative, but may produce invalid responses, prediction bias, class collapse, or confident incorrect predictions when applied to aerial imagery.

This project addresses these issues through a controlled comparison of multiple YOLO generations, RF-DETR, and four VLMs, examining both classification performance and model behaviour.

1.3 Project Objectives

ID	Objective
O1	Build standardized RGB/thermal aerial datasets from multiple sources.
O2	Train and evaluate YOLOv8, YOLOv10, YOLO11, YOLO12, and RF-DETR under a common framework.
O3	Evaluate InternVL2, BLIP-2, Qwen2.5-VL, and DeepSeek-VL for zero-shot aerial-object classification.
O4	Establish a unified evaluation framework using accuracy, precision, recall, F1, Jaccard, and related metrics.
O5	Compare model performance across RGB and thermal imagery.
O6	Analyse failure modes including prediction frequency, class collapse, invalid outputs, and per-class performance.
O7	Identify high-confidence incorrect predictions across selected confidence thresholds.
O8	Design and deliver a reproducible workflow and deployable tool for inference, metric calculation, analysis, and reporting.

Table 1.2: Objectives of the project

1.4 Research Questions

1. How does detection performance compare across YOLO generations?
2. How does RF-DETR compare with YOLO under the experimental framework?
3. How effectively can VLMs classify aerial objects without task-specific training?
4. How do InternVL2, BLIP-2, Qwen2.5-VL, and DeepSeek-VL differ in performance and prediction behaviour?
5. How does model performance change between RGB and thermal imagery?

6. Do the evaluated VLMs exhibit prediction collapse, invalid outputs, or high-confidence errors?
7. Which detector, VLM, and modality combination provides the most reliable performance within the evaluated setting?

1.5 Research Contributions

1. A unified dataset pipeline with standardized classes and annotations.
2. Comparative evaluation of YOLOv8, YOLOv10, YOLO11, YOLO12, and RF-DETR.
3. Comparative zero-shot evaluation of InternVL2, BLIP-2, Qwen2.5-VL, and DeepSeek-VL.
4. Cross-modality performance evaluation using RGB and thermal imagery.
5. A standardized metric framework including Micro/Macro F1 and Jaccard index.
6. Detailed VLM behaviour analysis including prediction frequency, class bias, entropy, invalid-output rate, and per-class recall.
7. High-confidence error analysis.
8. A reproducible and documented experimental workflow, delivered as a locally deployable inference tool (Chapter 6, Section 6.7).

1.6 Project Scope

The project focuses on vision-based aerial-object detection and classification using RGB and thermal imagery. Supervised object detectors are evaluated as the primary detection models, while VLMs are evaluated as zero-shot classification alternatives.

Detectors: YOLOv8, YOLOv10, YOLO11, YOLO12, RF-DETR.

VLMs: InternVL2-8B, BLIP-2, Qwen2.5-VL, DeepSeek-VL.

The target classes for supervised detection are Drone, Bird, Airplane, and Helicopter. The VLM evaluation additionally includes a *None* category to represent images without a target object.

Radio-frequency sensing, radar integration, and physical sensor fusion are outside the scope of the current experimental study.

1.7 Experimental Progress

This report documents the completed experimental workflow, including:

1. Dataset collection and consolidation.
2. Annotation conversion and class standardization.
3. Dataset cleaning and train/validation/test splitting.
4. A pilot study comparing five detector architectures to select the primary RGB and IR detectors.
5. Full-dataset training of the selected detectors (YOLOv12x for RGB, YOLOv10x for IR).
6. Preparation of RGB and thermal evaluation sets.
7. Zero-shot evaluation of InternVL2, BLIP-2, Qwen2.5-VL, and DeepSeek-VL.

8. Metric calculation and comparative analysis.
9. Prediction-distribution and class-collapse analysis.
10. Invalid-output analysis and high-confidence error analysis.
11. Design and implementation of a deployable inference and reporting tool (SkyEye) integrating both model families under a single interface.

Dataset construction is described in Chapter 5. Implementation, training, and the deployed tool are described in Chapter 6. Experimental results are presented in Chapter 7.

1.8 Scope Changes

The project originally considered a broader counter-UAV architecture including RF monitoring. The scope was subsequently narrowed to the models and modalities that form the final experimental comparison. RF sensing was excluded from the final study; all four VLM candidates – InternVL2, BLIP-2, Qwen2.5-VL, and DeepSeek-VL – were retained through to the final evaluation. The resulting study focuses on RGB and thermal imagery using five supervised detectors and four VLMs.

In addition, the detector architecture originally assumed as the primary RGB/IR model (YOLO11) was superseded during implementation: a pilot study (Chapter 6, Section 6.4) compared five detector architectures on a controlled sample and selected YOLOv12x for RGB and YOLOv10x for IR as the stronger performers, and these were the architectures carried forward to full-dataset training.

1.9 Document Structure

The remainder of this report is organized as follows.

Chapter 2 introduces the background concepts relevant to airborne-object detection, including RGB and thermal sensing, deep-learning detectors, and Vision-Language Models.

Chapter 3 reviews relevant datasets, object-detection architectures, and VLMs that form the basis of the experimental study.

Chapter 4 presents the experimental methodology, including supervised detection, zero-shot VLM evaluation, and the comparative evaluation procedure.

Chapter 5 describes the construction of the unified dataset, including source datasets, class standardization, annotation processing, preprocessing, and train, validation, and test splits.

Chapter 6 describes the experimental environment, hardware and software configuration, the pilot study and detector architecture selection, full-dataset training, VLM inference and prompting procedures, and the deployed SkyEye tool.

Chapter 7 presents the evaluation framework and experimental results for YOLOv8, YOLOv10, YOLO11, YOLO12, RF-DETR, InternVL2, BLIP-2, Qwen2.5-VL, and DeepSeek-VL.

Chapter 8 discusses the limitations of the study and the principal threats to the validity and generalizability of the reported results.

Chapter 9 summarizes the findings of the project and presents directions for future work.

2. Background

The detection and classification of low-altitude aerial objects has become an important research area due to the rapid proliferation of unmanned aerial vehicles (UAVs) in both civilian and military environments. Modern counter-UAV systems employ multiple sensing modalities to improve detection reliability under varying environmental conditions. This chapter introduces the fundamental technologies used throughout this project, including RGB vision systems, thermal imaging, deep learning-based object detection, Vision-Language Models (VLMs), and multimodal perception. These concepts provide the foundation for the methodology presented in later chapters.

2.1 Counter-UAV Sensing Technologies

Counter-UAV systems employ different sensing modalities depending on the operational environment and mission requirements. Radar remains the primary long-range sensing technology, while optical and thermal cameras are widely used for visual confirmation and object classification. Recent advances in deep learning have enabled vision-based systems to achieve high detection accuracy, particularly when multiple sensing modalities are combined.

Among the available sensing technologies, this project focuses on RGB and thermal imaging because they provide complementary information and are suitable for integration with deep learning models.

2.2 RGB Vision Systems

RGB cameras capture reflected visible light using red, green, and blue colour channels. Due to their relatively low cost, high spatial resolution, and ease of deployment, RGB cameras have become the dominant sensing modality for vision-based UAV detection systems.

When combined with convolutional neural networks such as YOLO, RGB imagery provides highly detailed object appearance information that enables accurate localization and classification. However, RGB cameras are sensitive to changes in illumination, shadows, fog, rain, and night-time conditions, which can significantly degrade detection performance.

Advantages	Limitations
High spatial resolution	Performance degrades at night
Low-cost hardware	Sensitive to fog, rain and haze
Rich colour and texture information	Affected by illumination changes
Suitable for real-time object detection	Small distant objects remain challenging

Table 2.1: Advantages and limitations of RGB vision systems

2.3 Thermal Imaging

Thermal cameras detect long-wave infrared (LWIR) radiation emitted by objects instead of reflected visible light. Since every object above absolute zero emits thermal radiation, thermal imaging enables object detection even in complete darkness.

Thermal imagery provides strong robustness against low-light environments, smoke, and moderate weather conditions, making it particularly useful for night-time surveillance. However, thermal sensors generally provide lower spatial resolution than RGB cameras, resulting in reduced texture information that can negatively affect fine-grained object classification.

Advantages	Limitations
Operates in complete darkness	Lower image resolution
Robust to illumination changes	Higher sensor cost
Insensitive to shadows	Limited colour and texture information
Suitable for adverse weather	Lower visual detail for classification

Table 2.2: Advantages and limitations of thermal imaging

2.4 Deep Learning Object Detection

Object detection combines object localization and classification within a single framework. Modern deep learning detectors predict both the object class and the corresponding bounding box simultaneously.

Deep learning detectors are generally categorized into two-stage detectors, such as Faster R-CNN, and one-stage detectors, such as the YOLO family. Two-stage detectors typically achieve high accuracy but require greater computational resources, whereas one-stage detectors prioritize real-time performance while maintaining competitive detection accuracy.

2.5 YOLO Object Detection

The You Only Look Once (YOLO) family has become one of the most widely adopted object detection frameworks because of its balance between detection accuracy and inference speed. YOLO performs object localization and classification in a single forward pass, enabling real-time operation on modern GPUs.

This project evaluates four successive YOLO generations – YOLOv8, YOLOv10, YOLO11, and YOLOv12 – alongside RF-DETR as a transformer-based alternative, for detecting four aerial object categories:

- Drone
- Bird
- Airplane
- Helicopter

Rather than assuming any single generation is best suited to both sensing modalities, the specific architecture used as the primary RGB and IR detector is determined empirically via a pilot study (Chapter 6, Section 6.4). YOLO-family detectors provide the initial detection stage of the proposed pipeline, before higher-level semantic reasoning is performed by Vision-Language Models when detector confidence is low.

2.6 Vision-Language Models

Vision-Language Models (VLMs) jointly process images and natural language, allowing them to perform zero-shot image understanding without task-specific training. Unlike conventional classifiers that are limited to predefined classes, VLMs can reason about previously unseen objects using textual prompts.

In this work, four VLMs are evaluated as semantic fallback models to analyze detected aerial objects and compare their classification capabilities across RGB and thermal imagery.

2.7 Multimodal Perception

RGB and thermal sensors provide complementary information. RGB images contain rich colour and texture information during favourable lighting conditions, whereas thermal imagery provides robust target visibility during night-time or adverse weather conditions.

Combining multiple sensing modalities can significantly improve detection robustness by compensating for the weaknesses of individual sensors. The overall architecture proposed in this project leverages both RGB and thermal imagery together with deep learning models to improve aerial object detection.

2.8 Evaluation Metrics

To evaluate the performance of the proposed system, several standard classification and object detection metrics are employed throughout this report. These include:

- Accuracy
- Precision
- Recall
- Intersection over Union (IoU)
- Mean Average Precision (mAP)
- Micro and Macro F1-score
- Micro and Macro Jaccard Index
- Per-class Recall
- Prediction Entropy
- Invalid Output Rate

A detailed mathematical description of each metric is presented in Chapter 7.

2.9 Comparative Summary

Modality	**Operating Conditions**	**Strengths**	**Primary Limitations**
Radar	Long-range	Long detection range	Limited performance for small UAVs
RGB	Daylight	High spatial detail	Sensitive to illumination and weather
Thermal	Day and night	Night-time operation	Lower spatial resolution

Table 2.3: Comparison of sensing modalities used in modern counter-UAV systems

This chapter introduced the fundamental concepts required to understand the remainder of this report. The following chapter reviews recent research on drone detection datasets, YOLO-based object detection, Vision-Language Models, and multimodal aerial surveillance systems.

3. Literature Review

This chapter reviews prior work relevant to the experimental study, focusing on airborne-object datasets, supervised object detectors, and Vision-Language Models (VLMs), and establishes the basis for the comparative RGB–thermal evaluation conducted in this project.

3.1 Airborne Object Detection Datasets

Public airborne-object datasets vary in class definitions, image sources, annotation formats, and sensing modality. This project draws on AOD-4, DUT-Anti-UAV, Drone-vs-Bird, and the Air Vehicles dataset.

AOD-4 provides multi-class airborne imagery spanning airplane, helicopter, drone, and bird categories [1]. DUT-Anti-UAV contributes UAV imagery for vision-based anti-UAV detection and tracking [2]. Drone-vs-Bird supplies examples for distinguishing UAVs from visually similar birds [3], while the Air Vehicles dataset adds further airborne-object diversity [4]. These sources were consolidated into a common four-class taxonomy for the detection experiments.

Dataset	**Modality**	**Primary Use**	**Main Contribution**
AOD-4 [1]	RGB	Detection	Multi-class airborne objects
DUT-Anti-UAV [2]	RGB/IR	UAV detection/tracking	UAV imagery and annotations
Drone-vs-Bird [3]	RGB	Object discrimination	Drone and bird examples
Air Vehicles [4]	RGB/IR	Airborne-object detection	Additional visual diversity

Table 3.1: Datasets contributing to the project dataset

Combining multiple sources increases variation in object appearance, background, scale, and acquisition conditions, requiring consistent class mapping and annotation standardization.

3.2 Deep-Learning Object Detectors

Single-stage detectors such as the YOLO family are widely used where high detection speed is required. This project evaluates YOLOv8, YOLOv10, YOLO11, and YOLO12 to compare successive generations of the architecture under a common task [5, 6, 7, 8], alongside RF-DETR as a transformer-based alternative [9].

All five detectors operate within the same four-class taxonomy — airplane, bird, drone, and helicopter — so their performance depends on the similarity between training distribution and evaluation targets.

Model	**Category**	**Role in Project**
YOLOv8 [5]	CNN-based detector	Comparative baseline
YOLOv10 [6]	Real-time detector	Comparative evaluation
YOLO11 [7]	Real-time detector	Comparative evaluation
YOLO12 [8]	Attention-based detector	Comparative evaluation
RF-DETR [9]	Transformer detector	Alternative architecture

Table 3.2: Supervised object detectors evaluated in this project

3.3 Vision-Language Models

VLMs combine visual representations with language processing to interpret images via natural-language instructions, and can be evaluated zero-shot without task-specific fine-tuning. InternVL targets large-scale multimodal tasks [?]; BLIP-2 connects pretrained visual encoders with large language models [10]; Qwen2.5-VL provides multimodal instruction-following and reasoning [11]; and DeepSeek-VL offers general-purpose vision-language understanding [12]. All four are evaluated under a common label space, prompting strategy, and evaluation procedure.

Model	**Category**	**Role in Project**
InternVL2 [?]	Vision-Language Model	Zero-shot multimodal reasoning
BLIP-2 [10]	Vision-Language Model	Zero-shot classification
Qwen2.5-VL [11]	Vision-Language Model	Zero-shot multimodal reasoning
DeepSeek-VL [12]	Vision-Language Model	Zero-shot visual classification

Table 3.3: Vision-Language Models evaluated in this project

3.4 RGB and Thermal Aerial Detection

RGB and thermal imagery are complementary: RGB offers richer colour and texture under adequate illumination, while thermal imagery yields useful target signatures under low-light conditions. Rather than fusing modalities during training, this project evaluates RGB and thermal data separately to isolate modality effects across detector and VLM architectures.

3.5 Research Gap

- Public datasets differ in class definitions, annotation formats, and sensing conditions, complicating direct comparison without standardization.
- Supervised detectors rely on a fixed vocabulary, requiring representative training data for reliable recognition.
- General-purpose VLMs offer zero-shot reasoning, but their reliability on airborne imagery across RGB and thermal modalities remains under-evaluated.

This project addresses these gaps through a unified four-class dataset and a controlled comparison of five detectors (YOLOv8, YOLOv10, YOLO11, YOLO12, RF-DETR) and four VLMs (InternVL2, BLIP-2, Qwen2.5-VL, DeepSeek-VL).

3.6 Chapter Summary

This chapter reviewed the datasets, detectors, and VLMs relevant to the study, establishing the need for consistent dataset preparation and controlled evaluation across architectures and modalities. Dataset construction is described in the following chapter.

4. Methodology

This chapter describes the methodology used to compare supervised object detectors and zero-shot Vision-Language Models (VLMs) for airborne-object recognition. Five detectors (YOLOv8, YOLOv10, YOLO11, YOLO12, RF-DETR) and four VLMs (InternVL2, BLIP-2, Qwen2.5-VL, DeepSeek-VL) were evaluated on standardized RGB and thermal datasets under a consistent experimental protocol.

4.1 Overall Experimental Framework

The study consists of two tracks: supervised detection, evaluating five detectors trained on the project dataset, and zero-shot VLM classification, evaluating four pretrained models without task-specific fine-tuning.

Component	**Models**	**Evaluation**
Object Detection	YOLOv8, YOLOv10, YOLO11, YOLO12	Supervised detection
Object Detection	RF-DETR	Supervised detection
VLM Classification	InternVL2, BLIP-2, Qwen2.5-VL, DeepSeek-VL	Zero-shot classification

Table 4.1: Models evaluated in the experimental framework

4.2 Supervised Object Detection

Detectors were trained and evaluated on the standardized four-class taxonomy: Airplane, Bird, Drone, Helicopter. Each produces bounding boxes, class labels, and confidence scores, assessed via precision, recall, mAP@0.50, mAP@0.50:0.95, and class-wise performance. Rather than assuming a single architecture is optimal for both sensing modalities, a pilot study (Chapter 6, Section 6.4) is used to select the strongest architecture per modality before committing to full-dataset training.

4.3 YOLO-Based Detectors

Four YOLO generations — YOLOv8, YOLOv10, YOLO11, YOLO12 — were trained under identical class definitions and dataset splits, enabling generational comparison for airborne-object detection.

Model	Architecture	Role
YOLOv8	Single-stage detector	Baseline comparison
YOLOv10	Single-stage detector	Generational comparison
YOLO11	Single-stage detector	Generational comparison
YOLO12	Single-stage detector	Generational comparison

Table 4.2: YOLO detectors evaluated in the study

4.4 RF-DETR

RF-DETR, a transformer-based detector, was trained and evaluated under the same four-class vocabulary and datasets, with results compared against the YOLO models using identical metrics.

4.5 RGB and Thermal Evaluation

RGB and thermal modalities are evaluated separately, as they capture different visual information under different environmental conditions.

Modality	Main Information	Primary Challenge
RGB	Colour, texture, spatial detail	Illumination, weather, clutter
Thermal	Thermal radiation, heat signatures	Lower spatial/textural detail

Table 4.3: Characteristics of the evaluated sensing modalities

4.6 Vision-Language Model Evaluation

InternVL2, BLIP-2, Qwen2.5-VL, and DeepSeek-VL were evaluated zero-shot, with no task-specific fine-tuning, using identical image sets and a standardized prompt. The VLM task uses a five-class taxonomy — Airplane, Bird, Drone, Helicopter, and None — where *None* allows a model to indicate no target is present, unlike the four-class supervised-detector taxonomy.

4.7 Prompting and Output Processing

A single fixed prompt was applied across all four VLMs to minimize formulation-driven variation. Responses unmappable to one of the five valid classes were recorded as invalid outputs rather than forced into an arbitrary category, and the invalid-output rate is reported separately.

4.8 VLM Evaluation Protocol

- **Label space:** Airplane, Bird, Drone, Helicopter, None.
- **Prompt:** One fixed classification prompt for all models.
- **Evaluation data:** Identical RGB/thermal image sets across models.
- **Training:** No task-specific fine-tuning.
- **Output handling:** Invalid/unmappable responses recorded separately.

4.9 Evaluation Metrics

Detectors: Precision, Recall, mAP@0.50, mAP@0.50:0.95, per-class performance, confusion matrices.

VLMs: Accuracy, Precision, Recall, Micro/Macro F1, per-class F1, Micro/Macro/per-class Jaccard, invalid-output rate, and prediction frequency (to identify class bias or collapse).

4.10 Confidence and Fallback Concept

The design uses a hybrid detector–VLM pipeline: a supervised detector acts as the primary layer, with VLM fallback triggered on low-confidence detections:

$$\text{Input Image} \rightarrow \text{Primary Detector} \rightarrow \text{Confidence Assessment} \rightarrow \begin{cases} \text{Accept Detection}, & C \geq \tau \\ \text{VLM Analysis}, & C < \tau \end{cases} \tag{4.1}$$

where τ is a tunable threshold.

This design has since been implemented as the **Adaptive Fallback** pipeline mode in the SkyEye tool (Chapter 6, Section 6.7): the primary detector runs first on every image, and any detection whose confidence falls below a user-configurable cutoff is re-classified by the selected VLM, which becomes the source of truth for that result. The threshold τ is exposed directly in the tool's UI as the confidence cutoff rather than fixed at a hand-set default. Two additional operating modes are also available and do not use this cascade: **YOLO Only**, which runs detection alone at minimum latency, and **VLM Only**, which routes every image to the VLM regardless of detector confidence for maximum contextual accuracy at the cost of latency.

Systematic calibration of τ against the full-dataset detector confidence distributions (Chapter 7, Section 7.3) – rather than a hand-set default – remains future work (Chapter 8, Section 8.8).

4.11 Comparative Analysis

Detectors are compared across YOLO generations and against RF-DETR; VLMs are compared under the same zero-shot protocol. All results are split by RGB and thermal modality, with class-wise metrics examined alongside aggregate scores so strong overall performance does not mask weaknesses on individual categories.

4.12 Experimental Workflow

1. Construct and standardize RGB and thermal datasets.
2. Define the four-class detection vocabulary.
3. Run a pilot study across five detector architectures and select the strongest per modality.
4. Train the selected architectures (YOLOv12x RGB, YOLOv10x IR) on the full dataset.
5. Train and evaluate RF-DETR alongside the pilot comparison.
6. Define the five-class VLM vocabulary including None.
7. Evaluate InternVL2, BLIP-2, Qwen2.5-VL, and DeepSeek-VL with the fixed prompt.
8. Calculate model-specific evaluation metrics.
9. Analyse per-class performance and prediction distributions.
10. Measure invalid VLM outputs and high-confidence errors.
11. Compare performance across RGB and thermal modalities.
12. Integrate both model families into a deployable tool with a confidence-gated fallback pipeline.

4.13 Chapter Summary

This chapter described the methodology for comparing five supervised detectors and four zero-shot VLMs across RGB and thermal datasets, using a four-class taxonomy for detection and a five-class taxonomy (including None) for VLMs, under a common evaluation protocol, together with the confidence-gated fallback design later implemented in the deployed tool. Experimental findings follow in the next chapter.

5. Dataset Construction

The performance of an aerial-object detection system depends strongly on the quality, diversity, and consistency of its training and evaluation data. A dedicated dataset-construction pipeline was therefore developed for this project, combining multiple public aerial-object datasets, converting their annotations to a common format, standardizing class labels, and organizing the result into consistent training, validation, and test splits.

Separate RGB and infrared (IR) datasets were constructed and retained independently, enabling modality-specific training and evaluation across the detectors and Vision-Language Models investigated in this work.

5.1 Dataset Sources

The RGB dataset combines three public sources: the Drone vs Bird dataset (Kaggle), the Airborne Object Detection dataset AOD-4 (Roboflow), and DUT-Anti-UAV (GitHub). The IR dataset combines two public sources: the Air Vehicles dataset (Roboflow) and Anti-UAV410 (GitHub).

Dataset	**Modality**	**Task**	**Purpose in This Project**
Drone vs Bird	RGB	Classification/Detection	Drone/bird appearance diversity
AOD-4	RGB	Object Detection	Multi-class aerial-object detection
DUT-Anti-UAV	RGB	Detection/Tracking	UAV imagery under challenging conditions
Air Vehicles	IR	Detection	Thermal aerial-vehicle imagery
Anti-UAV410	IR	Tracking	Large-scale thermal UAV benchmark (JSON-annotated)

Table 5.1: Public data sources used for dataset construction

Combining multiple sources increased variation in object appearance, scale, viewpoint, background, and illumination, while reducing dependence on the characteristics of any single dataset.

5.2 Target Class Definition

Source datasets used differing annotation schemes, so the detection task was standardized around four airborne-object categories. Source labels were mapped to the unified class IDs shown below.

Class ID	Final Class	Description
0	Airplane	Fixed-wing aircraft
1	Bird	Bird or bird-like airborne target
2	Drone	Unmanned aerial vehicle
3	Helicopter	Rotary-wing aircraft

Table 5.2: Unified class definition used for object detection

5.3 Annotation Conversion

All annotations were converted into a common YOLO11-compatible format. Each object annotation is stored as

$$(c, x_c, y_c, w, h) \tag{5.1}$$

where c is the class identifier and (x_c, y_c, w, h) are the normalized bounding-box centre coordinates, width, and height. Anti-UAV410, originally distributed as video sequences with JSON annotations, required explicit conversion to YOLO TXT label files before merging with the other IR source.

5.4 RGB Dataset: Cleaning and Verification

The merged raw RGB dataset initially showed a mismatch between image and label counts in every split, indicating unannotated images requiring removal.

Split	Images	Labels
Train	10,158	9,979
Validation	2,998	2,936
Test	1,500	1,321
Total	14,656	14,236

Table 5.3: Raw RGB dataset file counts before cleaning

Cleaning consisted of: annotation-format verification; removal of images lacking a label file; a file-extension audit (confirming only `.jpg`/`.jpeg` files); image-corruption and resolution checks (145×177 to 4032×5616 px, no corrupted files found); an empty-label-file check (none found); per-annotation validation of class IDs, normalized coordinates, and box bounds; and a final image–label correspondence check.

Split	Raw	Final	Removed	% Removed
Train	10,158	9,978	180	1.77%
Validation	2,998	2,936	62	2.07%
Test	1,500	1,321	179	11.93%
Total	14,656	14,235	421	2.87%

Table 5.4: RGB dataset size before and after cleaning

All 421 removed images (2.87% of the raw total) lacked a corresponding annotation file; no images were removed for corruption, empty labels, or unsupported format. Class balance was not artificially adjusted, so the final distribution reflects the natural composition of the merged sources.

5.5 IR Dataset Construction

The Air Vehicles dataset was originally split approximately 78%/14%/8% (train/validation/test) and was reorganized to a standard 70%/20%/10% split for consistency before merging with the converted Anti-UAV410 data. The combined dataset was then reduced while preserving class diversity and balance, yielding a final IR dataset of approximately 15,552 images.

5.6 Final Dataset Distribution

Class	Train	Validation	Test	Total
Airplane	1,273	366	180	1,819
Bird	1,697	337	127	2,161
Drone	5,068	1,750	774	7,592
Helicopter	1,940	483	240	2,663
Total	9,978	2,936	1,321	14,235

Table 5.5: Final RGB dataset distribution

Class	Train	Validation	Test	Total
Airplane	1,900	543	272	2,715
Bird	2,030	580	290	2,900
Drone	5,569	1,591	796	7,956
Helicopter	1,387	396	198	1,981
Total	10,886	3,110	1,556	15,552

Table 5.6: Final infrared (IR) dataset distribution

Across both modalities, Drone is the dominant class (7,592 RGB / 7,956 IR instances) while Helicopter and Airplane are comparatively underrepresented in IR and RGB respectively — an imbalance revisited during model evaluation in Chapter 7.

5.7 Dataset Construction Pipeline

1. Collection of images and annotations from multiple public sources.
2. Separation of RGB and IR imagery.
3. Conversion of annotations (including JSON-to-YOLO for Anti-UAV410) to a common format.
4. Mapping of source labels to the unified four-class vocabulary.
5. Removal of unlabeled, corrupted, or invalid samples.
6. Verification of image–annotation correspondence.
7. Re-splitting into consistent 70/20/10 train, validation, and test sets.
8. Calculation of final dataset statistics.
9. Generation of datasets for model training and evaluation.

This process produced standardized RGB (14,235 images) and IR (15,552 images) datasets supporting the comparison of YOLOv8, YOLOv10, YOLO11, YOLO12, RF-DETR, InternVL2, BLIP-2, Qwen2.5-VL, and DeepSeek-VL.

5.8 Dataset Limitations

Despite standardization, source datasets differ in camera characteristics, resolution, background, and annotation quality. The RGB dataset spans a wide resolution range (145×177 to 4032×5616 px), reflecting mixed source origins. IR data are less diverse than RGB, and class balance was preserved rather than enforced, leaving Drone overrepresented relative to other classes. These factors may introduce dataset bias affecting model generalization, discussed further in Chapter 8. In particular, the comparatively narrower diversity of the IR sources (Section 5.1) is identified as the leading cause of the IR detector's lower full-dataset performance relative to RGB (Chapter 7, Section 7.3.5), rather than a limitation of the YOLOv10x architecture itself.

5.9 Chapter Summary

This chapter described the construction of the standardized RGB and IR datasets used throughout the project. Three RGB sources (Drone vs Bird, AOD-4, DUT-Anti-UAV) and two IR sources (Air Vehicles, Anti-UAV410) were combined, labels mapped to a common four-class vocabulary, annotations converted to a consistent YOLO format, and invalid samples removed through systematic quality control — yielding 14,235 RGB and 15,552 IR images across training, validation, and test splits, forming the common experimental basis for the comparisons presented in the following chapters.

6. Implementation

This chapter describes what has actually been built and configured to date, as distinct from the target design described in Chapter 4.

6.1 Software and Hardware Stack

All experiments in this project – the pilot detector study, the VLM zero-shot evaluation, and the full-dataset detector training – were run locally on a single machine allocated by the university, rather than on a cloud platform. Table 6.1 summarizes the environment.

Component	**Specification**
Operating System	Ubuntu 24.04
GPU	1× NVIDIA RTX 4000 Ada Generation (20 GB VRAM)
System RAM	32 GB
CUDA	CUDA 12.1
PyTorch	2.5.1+cu121 (CUDA available: True), cuDNN 90100
Detection Frameworks	Ultralytics (YOLOv8x, YOLOv10x, YOLOv12x), RF-DETR
VLMs Evaluated	InternVL2-8B, DeepSeek-VL-7B-Chat, BLIP-2, Qwen2.5-VL-7B
Deployment Interface	Streamlit-based web tool, SkyEye (Section 6.7)
Experiment Tracking	Local logging with structured CSV/JSON output

Table 6.1: Hardware and software stack used in the implementation

Note: An earlier version of this chapter referenced Grounding DINO as a candidate open-vocabulary detector and YOLOv11x as the trained primary RGB detector, run on a Kaggle dual-Tesla-T4 platform. Neither reflects the current experimental design: Grounding DINO was never run as part of this project, and YOLOv11x was superseded after the pilot study (Section 6.4) identified YOLOv12x and YOLOv10x as stronger candidates for RGB and IR respectively. Both references have been removed; the current primary detectors are YOLOv12x (RGB) and YOLOv10x (IR), trained locally on the hardware in Table 6.1 using the full dataset (Section 6.5).

6.2 Dataset Preparation

Training data is drawn from the standardized RGB and IR datasets described in Chapter 5. Table 6.2 restates the final split sizes used for both the pilot study sampling and the full-dataset training runs.

Dataset	Train	Validation	Test	Total
RGB	9,978	2,936	1,321	14,235
IR (Thermal)	10,886	3,110	1,556	15,552

Table 6.2: Final dataset split used for training, validation, and testing.

Class	Train	Validation	Test	Total
Airplane	1,273	366	180	1,819
Bird	1,697	337	127	2,161
Drone	5,068	1,750	774	7,592
Helicopter	1,940	483	240	2,663
Total	**9,978**	**2,936**	**1,321**	**14,235**

Table 6.3: Final RGB dataset distribution by class.

Class	Train	Validation	Test	Total
Airplane	1,900	543	272	2,715
Bird	2,030	580	290	2,900
Drone	5,569	1,591	796	7,956
Helicopter	1,387	396	198	1,981
Total	**10,886**	**3,110**	**1,556**	**15,552**

Table 6.4: Class-wise distribution of the IR (thermal) dataset.

Split	**Raw Images**	**Final Images**	**Removed**	**% Removed**
Train	10,158	9,978	180	1.77%
Validation	2,998	2,936	62	2.07%
Test	1,500	1,321	179	11.93%
Total	**14,656**	**14,235**	**421**	**2.87%**

Table 6.5: RGB dataset size before and after dataset cleaning. Duplicate, corrupted, and unsuitable images were removed prior to training.

Cleaning methodology (annotation-format verification, image–label correspondence checks, corruption/resolution audits) is unchanged from Chapter 5 and is not repeated here.

6.3 Annotation Strategy

Background categories are planned to be annotated using blank label files rather than being omitted from the training set entirely, so that the detector learns to actively suppress these categories rather than simply never encountering them. This strategy is designed but not yet applied to the full dataset (see F4, Chapter 9, Section 9.2).

6.4 Pilot Study: Detector Architecture Selection

Before committing GPU time to full-dataset training, a pilot study was conducted to compare candidate detector architectures on a controlled sample of the dataset and select the strongest performer per modality, rather than assuming a single architecture (e.g. the most recent YOLO release) would be optimal for both RGB and IR.

6.4.1 Pilot Setup

Five architectures – YOLOv8x, YOLOv10x, YOLO11x, YOLOv12x, and RF-DETR – were trained and evaluated locally under identical hardware, software, and hyperparameter conditions, isolating architectural differences from confounding factors such as differing compute environments or training schedules.

Parameter	**Value**
Epochs	100
Image size (imgsz)	616
Batch size	8
Optimizer	AdamW
Initial learning rate	0.001
Patience	20

Table 6.6: Pilot study training configuration

Modality	Images	Instances
RGB	310	389
IR	310	373

Table 6.7: Pilot study evaluation split

The same image and instance counts were held constant across every model/modality combination to prevent data-split bias from confounding the architecture comparison.

6.4.2 Pilot Results – Overall Metrics

Model	Modality	P	R	F1	mAP50	mAP50-95
YOLOv8x	RGB	0.864	0.772	0.815	0.809	0.463
YOLOv8x	IR	0.598	0.567	0.582	0.566	0.244
YOLOv10x	RGB	0.595	0.558	0.576	0.571	0.304
YOLOv10x	IR	0.665	0.584	**0.622**	0.565	**0.276**
YOLO11x	RGB	N/A	0.820*	0.530†	0.533	N/A
YOLO11x	IR	0.676	0.554	0.609	0.574	0.250
YOLOv12x	RGB	0.779	0.804	0.791	**0.849**	0.472
YOLOv12x	IR	0.589	0.548	0.568	0.499	0.214
RF-DETR	RGB	0.884	0.756	0.811	0.821	**0.475**
RF-DETR	IR	0.681	0.585	0.580	0.493	0.206

Table 6.8: Pilot study overall metrics by model and modality. *Maximum recall derived from the P/R curve. †Peak F1 derived from the F1-confidence curve.

6.4.3 Architecture Selection

RGB – YOLOv12x selected. YOLOv12x achieved the highest overall mAP50 (0.849) of the five candidates, indicating the strongest bounding-box localization, together with a high overall recall (0.804). RF-DETR was competitive on the stricter mAP50-95 metric (0.475 vs. 0.472), but YOLOv12x's clear lead on mAP50 and its lower inference cost as a single-stage detector made it the preferred primary RGB detector going forward.

IR – YOLOv10x selected. Thermal imagery inherently lacks the spatial and textural resolution of RGB, making tight localization harder for all models – reflected in the uniformly lower mAP50-95 scores across the IR column. YOLOv10x achieved the highest strict-accuracy score (mAP50-95 = 0.276) and the highest overall F1-score (0.622) of the five candidates, indicating the best balance of minimizing false positives while maximizing detection under thermal conditions.

Modality	Selected Model	Basis
RGB	YOLOv12x	Highest mAP50 (0.849), strong recall (0.804)
IR (Thermal)	YOLOv10x	Highest mAP50-95 (0.276), highest F1 (0.622)

Table 6.9: Pilot study – selected architectures

Model	Modality	Class	P	R	F1	mAP50	mAP50-95
YOLOv12x	RGB	Airplane	0.767	0.876	0.818	0.840	0.591
YOLOv12x	RGB	Bird	0.708	0.761	0.734	0.813	0.340
YOLOv12x	RGB	Drone	0.785	0.775	0.780	0.804	0.431
YOLOv12x	RGB	Helicopter	0.854	0.839	0.846	0.904	0.526
YOLOv10x	IR	Airplane	0.646	0.681	0.663	0.660	0.317
YOLOv10x	IR	Bird	0.413	0.544	0.470	0.413	0.161
YOLOv10x	IR	Drone	0.867	0.321	0.469	0.381	0.223
YOLOv10x	IR	Helicopter	0.735	0.791	0.762	0.805	0.404

Table 6.10: Pilot study – per-class metrics for selected architectures

Notably, YOLOv10x on IR shows a large precision–recall gap on Drone (0.867 precision vs. 0.321 recall) even at pilot scale – consistent with the general observation elsewhere in this report that drone is the hardest class to detect reliably in thermal imagery, likely due to its small size and low thermal signature relative to engine-driven targets like helicopters and airplanes.

6.5 Full-Dataset Detector Training

Following the pilot study, full-dataset training was carried out for the two selected architectures on the same hardware described in Table 6.1:

- **YOLOv12x** on the full RGB dataset (9,978 train / 2,936 validation images, Table 6.3)
- **YOLOv10x** on the full IR dataset (10,886 train / 3,110 validation images, Table 6.4)

Both runs have now produced usable checkpoints. YOLOv12x completed a single, uninterrupted 100-epoch run; YOLOv10x's training was interrupted and resumed across four separate runs, accumulating 59 epochs in total. The two training executions are summarized in Table 6.11, and full quantitative results are presented in Chapter 7, Section 7.3.

Characteristic	YOLOv12x (RGB)	YOLOv10x (IR)
Input resolution	768 px	768 px
Batch size	4	4
Total epochs	100	59
Number of runs	1 (uninterrupted)	4 (interrupted & resumed)
Total training time	88,948.6 s (≈24.71 h)	32,315.1 s (≈8.98 h)
Average time per epoch	≈889 s (≈14.8 min)	≈547 s (≈9.1 min)
GPU	NVIDIA RTX 4000 Ada Generation	

Table 6.11: Full-dataset training configuration and execution for the two primary detectors

Both detectors were trained under identical input resolution (768 px) and batch size (4) settings, so any difference in per-epoch training time reflects factors other than input resolution. YOLOv10x's shorter total training time is not directly comparable to YOLOv12x's, since YOLOv10x accumulated only 59 epochs across four interrupted runs, whereas YOLOv12x completed a single uninterrupted 100-epoch run (full results reported in Chapter 7, Section 7.3).

For YOLOv10x, the reported benchmark corresponds to its highest-performing checkpoint, observed around epochs 57–59; the optimal checkpoint varies slightly depending on which metric (precision/recall/F1 vs. mAP50 vs. mAP50-95) is prioritized (Chapter 7, Section 7.3.3).

6.6 Vision-Language Model Evaluation

In parallel with detector training, four pretrained VLMs – InternVL2-8B, DeepSeek-VL-7B-Chat, BLIP-2, and Qwen2.5-VL-7B – were evaluated zero-shot on the full RGB and IR datasets, without any task-specific fine-tuning, to assess their viability as a fallback classification layer and as a standalone option in the deployed tool (Section 6.7).

6.6.1 Evaluation Protocol

All four models were evaluated under an identical protocol to ensure a fair comparison:

- **Label space:** a five-class taxonomy – airplane, bird, drone, helicopter, and *none* (no object of interest / background) – rather than the four-class taxonomy used for the supervised detectors, allowing VLMs to explicitly signal the absence of a target.
- **Prompt:** a single fixed instruction prompt requesting a one-word class label, applied identically across all models and both modalities to minimize formulation-driven variation.
- **Evaluation sets:** the full IR set (10,886 images) and full RGB set (9,976 images), identical across all four models so that comparisons are not confounded by differing test data.

- **Output parsing:** raw text responses were parsed and matched against the five-class label set; responses that could not be matched to a valid label were logged as invalid rather than discarded or forced into a class.

Model	Approximate Size
InternVL2-8B	8B
DeepSeek-VL-7B-Chat	7B
Qwen2.5-VL-7B	7B
BLIP-2	2.7B

Table 6.12: VLMs evaluated in this project

For each model and modality, accuracy, precision/recall (micro and macro), micro/macro F1, micro/macro Jaccard index, mean inference latency, invalid-output rate, and prediction-frequency distribution were computed. Full results are presented in Chapter 7, Section 7.5.

6.7 Deployment Tool

The trained detectors and evaluated VLMs are integrated into **SkyEye**, a locally deployable web-based tool that lets a user upload imagery and run either model family, or both in cascade, through a single interface rather than only reading results from static offline experiments.

6.7.1 Architecture

SkyEye runs as a set of Docker Compose services, summarized below.

- `app` – the Streamlit UI and orchestration layer. It carries no GPU reservation itself, so `docker compose up` succeeds on any machine with or without a GPU; if a detection service is unreachable, the app reports this directly rather than returning a silent or fabricated result.
- `yolo` – a FastAPI service wrapping the Ultralytics YOLO framework. Both trained checkpoints (YOLOv12x for RGB, YOLOv10x for IR, Section 6.5) are loaded at startup and kept resident in VRAM. Class-agnostic non-maximum suppression is applied on top of Ultralytics' own per-class NMS, since a single real object can otherwise receive multiple overlapping boxes under different predicted classes.
- `vlm` – a single FastAPI service shared by the self-hosted VLMs (DeepSeek-VL-7B-Chat and BLIP-2; InternVL2 is implemented but currently disabled in the UI – see Section 6.7.4). Rather than keeping every model resident in VRAM simultaneously, which does not fit on a single consumer GPU, the service lazy-loads whichever model is requested and unloads the previous one on a switch. Model weights are pre-fetched to local disk at container startup so a later switch pays only the VRAM-load cost, not a fresh download.

- **Qwen2.5-VL** is called directly from `app` via the OpenRouter hosted API rather than through the local `vlm` service, since it is the one evaluated VLM with no local GPU footprint.
- `postgres` – stores every run (single image or folder batch): which files were processed, every detection (class, confidence, bounding box, latency, producing model), and whether a result was escalated from YOLO to a VLM.

Layer	**Technology**
UI	Streamlit
Object detection	Ultralytics YOLO (YOLOv12x RGB, YOLOv10x IR)
Vision-language models	InternVL2†, DeepSeek-VL-7B-Chat, BLIP-2 (self-hosted); Qwen2.5-VL (via OpenRouter)
Model serving	FastAPI + Uvicorn
Database	PostgreSQL 16 + SQLAlchemy
Deployment	Docker Compose
GPU inference	PyTorch + CUDA 12.1, `bitsandbytes` 8-bit quantization

Table 6.13: SkyEye tool stack. †Implemented but disabled in the shipped UI; see Section 6.7.4.

6.7.2 Pipeline Modes

The tool exposes three operating modes, corresponding directly to the confidence-gated fallback concept in Chapter 4, Section 4.8:

- **YOLO Only (EDGE-FAST)** – detection only, lowest latency, no VLM fallback.
- **VLM Only (DEEP-VLM)** – every image is routed to the selected VLM, maximizing contextual accuracy at the cost of latency.
- **YOLO & VLM (Adaptive Fallback)** – YOLO classifies every image first; any result below a user-configurable confidence cutoff is re-checked by the VLM, which becomes the source of truth for that detection.

6.7.3 Usage

The interface provides three views: **IMAGE**, for uploading and analyzing a single image under a chosen pipeline mode and confidence cutoff, with every YOLO-detected object individually boxed; **FOLDER**, for batch-processing a directory of images into a CSV of per-image results plus annotated output images; and **HISTORY**, for browsing and drilling into any past run and its recorded detections, persisted in PostgreSQL.

6.7.4 Known Limitations

InternVL2 is implemented within the shared `vlm` service but is currently disabled in the user interface, pending confirmation of the exact model build against the InternVL2-8B checkpoint evaluated in Section 7.5 and a stated resource/integration justification, to be recorded here in a future revision. A mock inference mode exists in the codebase to allow the rest of the application's logic to be exercised on a machine without a GPU, but it is opt-in only and is never enabled in the shipped deployment configuration; every model call in the deployed tool either returns a real inference result or reports clearly that the model is unavailable, rather than substituting a simulated or fabricated result.

7. Evaluation Framework and Results

7.1 Evaluation Metrics

Detector performance is assessed using the metrics defined in Table 7.1; VLM performance uses the classification-oriented metrics defined in Chapter 6, Section 6.6.1.

Metric	Description
Precision	Proportion of detections that are correct
Recall	Proportion of actual targets that are detected
F1 Score	Harmonic mean of precision and recall
mAP50	Mean average precision at 0.5 IoU threshold
mAP50-95	Mean average precision averaged over IoU thresholds 0.5–0.95
Invalid Output Rate	Proportion of VLM outputs unmappable to a valid class
Latency	Mean end-to-end processing time per image

Table 7.1: Evaluation metrics used to assess detection performance

7.2 Pilot Study Results

The pilot study comparing five detector architectures (YOLOv8x, YOLOv10x, YOLO11x, YOLOv12x, RF-DETR) on a controlled sample constitutes the first completed detector-evaluation results in this project. Full results, per-class breakdowns, and the resulting architecture selection are reported in Chapter 6, Section 6.4, and are not repeated here. In summary: **YOLOv12x** was selected as the primary RGB detector (mAP50 = 0.849) and **YOLOv10x** as the primary IR detector (mAP50-95 = 0.276, F1 = 0.622), and both were subsequently trained on the full dataset (Chapter 6, Section 6.5).

7.3 Full-Dataset Detector Results

Full-dataset training for YOLOv12x (RGB) and YOLOv10x (IR) has produced usable checkpoints, using the configurations summarized in Chapter 6, Table 6.11. Both models demonstrated substantial performance improvements over their respective pilot-study results (Section 7.3.4).

7.3.1 Combined Benchmark

Modality	Model	P	R	F1	mAP50	mAP50-95
RGB	YOLOv12x	**0.9329**	**0.9126**	**0.9226**	**0.9411**	**0.6466**
IR	YOLOv10x	0.7858	0.7410	0.7627	0.6657	0.3381

Table 7.2: Full-dataset primary detector results. RGB: YOLOv12x, 100 epochs (1 run). IR: YOLOv10x, 59 epochs (4 runs), best checkpoint ≈epoch 58.

The RGB detector reached 93.29% precision, 91.26% recall, 92.26% F1-score, 94.11% mAP50, and 64.66% mAP50-95. The thermal detector reached 78.58% precision, 74.10% recall, 76.27% F1-score, 66.57% mAP50, and 33.81% mAP50-95. For YOLOv10x, the reported metrics correspond to its highest-performing checkpoint, observed around epoch 58; the exact optimal epoch varies slightly by metric (Section 7.3.3).

7.3.2 YOLOv12x (RGB) Training Progression

The RGB detector showed rapid early improvement and continued, stable convergence through all 100 epochs. Validation classification loss fell from 2.1825 at epoch 1 to 0.6174 at epoch 100, and validation box loss fell from 2.2263 to 1.1805 over the same period.

Phase	Epoch	P	R	F1	mAP50	mAP50-95	Val Cls Loss
Initial	1	0.4398	0.4222	0.4308	0.3955	0.1672	2.1825
Early Stage	10	0.8059	0.7040	0.7515	0.7764	0.4255	1.2089
Quarter Stage	25	0.8605	0.8491	0.8548	0.8854	0.5325	0.9042
Midpoint	50	0.9187	0.8894	0.9038	0.9228	0.6045	0.7225
Late Stage	75	0.9286	0.9132	0.9208	0.9380	0.6356	0.6417
Final/Peak	**100**	**0.9329**	**0.9126**	**0.9226**	**0.9411**	**0.6466**	**0.6174**

Table 7.3: YOLOv12x (RGB) training progression at milestone epochs

7.3.3 YOLOv10x (IR) Training Progression

Thermal training was interrupted and resumed across four separate runs, accumulating 59 total epochs. The model improved consistently despite the interruptions; precision rose from 0.3674 to 0.7858 and recall from 0.3340 to 0.7410 between epoch 1 and the best checkpoint. Most notably, the low recall observed during the pilot study (0.5840) was substantially addressed, reaching 0.7410 at the F1-score peak.

Phase	**Epoch**	**P**	**R**	**F1**	**mAP50**
Initial	1	0.3674	0.3340	0.3499	0.2368
Early Stage	10	0.6440	0.6133	0.6283	0.5570
Midpoint	30	0.7155	0.6771	0.6958	0.6116
mAP50-95 Peak	57	0.7800	0.7378	0.7583	0.6643
F1-Score Peak	**58**	**0.7858**	**0.7410**	**0.7627**	0.6637
mAP50 Peak	59	0.7819	0.7395	0.7601	**0.6657**

Table 7.4: YOLOv10x (IR) training progression at milestone epochs. The optimal checkpoint varies slightly by metric: epoch 57 for mAP50-95, epoch 58 for precision/recall/F1, epoch 59 for mAP50.

7.3.4 Comparison Against the Pilot Study

Both detectors improved substantially between the pilot study (Chapter 6, Section 6.4, 310-image validation sample, 616 px/batch 8) and full-dataset training (768 px/batch 4, Chapter 6, Table 6.11), confirming that the architectures selected via the pilot benefited from the additional training data rather than having already saturated their capacity at pilot scale. Because the training resolution and batch size also changed between the two phases, the reported gains reflect the combined effect of more data and a revised training configuration, not data volume alone.

Metric	**Pilot (310 val. img.)**	**Full Dataset**	**Abs. Gain**	**Rel. Improvement**
Precision	0.7790	**0.9329**	+0.1539	**+19.76%**
Recall	0.8040	**0.9126**	+0.1086	**+13.51%**
F1-Score	0.7910	**0.9226**	+0.1316	**+16.64%**
mAP50	0.8490	**0.9411**	+0.0921	**+10.85%**
mAP50-95	0.4720	**0.6466**	+0.1746	**+36.99%**

Table 7.5: YOLOv12x (RGB): pilot vs. full-dataset performance

Metric	**Pilot (310 val. img.)**	**Full Dataset**	**Abs. Gain**	**Rel. Improvement**
Precision	0.6650	**0.7858**	+0.1208	**+18.17%**
Recall	0.5840	**0.7410**	+0.1570	**+26.88%**
F1-Score	0.6220	**0.7627**	+0.1407	**+22.62%**
mAP50	0.5650	**0.6657**	+0.1007	**+17.82%**
mAP50-95	0.2760	**0.3381**	+0.0621	**+22.50%**

Table 7.6: YOLOv10x (IR): pilot vs. full-dataset performance

For YOLOv12x, the largest relative gain was in mAP50-95 (+36.99%), indicating that additional training data disproportionately improved precise bounding-box localization rather than coarse detection. For YOLOv10x, the largest relative gain was in recall (+26.88%), directly addressing the low-recall limitation identified in the pilot study (Chapter 6, Table 6.8).

7.3.5 IR Performance Discussion

Relative to the RGB detector, YOLOv10x's full-dataset results (Table 7.2) are lower across all metrics. This gap is attributed primarily to the quality and consistency of the available IR source data rather than to a limitation of the YOLOv10x architecture itself. As noted in Chapter 5, Section 5.8, the IR dataset is less diverse than the RGB dataset and was assembled from fewer public sources, with less consistent annotation quality and narrower coverage of object appearance, scale, and background variation. Both detectors were trained under identical resolution (768 px) and batch size (4) settings (Chapter 6, Table 6.11), so this discrepancy is not attributable to differing training configurations. The recall improvement achieved between the pilot and full-dataset IR runs (Table 7.6) indicates that YOLOv10x continues to benefit from additional training data, and further gains would plausibly follow from a larger and more consistently annotated IR dataset (F1, Chapter 9, Section 9.2).

7.3.6 Summary

Both full-dataset detectors substantially outperformed their pilot-study counterparts and confirm the architecture selections made in the pilot (Chapter 6, Section 6.4). **YOLOv12x** attained strong performance on RGB imagery across all measured metrics. **YOLOv10x** showed comparatively lower but meaningfully improved performance on IR imagery, consistent with the more limited IR dataset available for training rather than an architectural shortcoming. Confusion matrices and precision/recall/F1-confidence curves for both full-dataset models are not yet available and are identified as an item for future reporting (Chapter 9, Section 9.2).

7.4 Experimental Scenarios

The system is intended to be evaluated across the operational scenarios listed in Table 7.7, designed to probe robustness across environmental conditions and target types. With full-dataset training for both primary detectors complete (Section 7.3) and the confidence-gated fallback pipeline now implemented in the SkyEye tool (Chapter 6, Section 6.7), scenario testing is no longer blocked by either dependency. No scenario runs have been recorded yet as of this writing; Table 7.7 also serves as the reporting template to be populated once testing begins.

No.	Scenario	Purpose
1	Day	Baseline performance under nominal lighting
2	Night	Evaluate thermal-channel reliance and RGB degradation
3	Fog	Evaluate robustness under reduced visibility
4	Novel drones	Evaluate fallback-layer generalization to unseen airframes
5	Bird interference	Evaluate false-positive suppression against biological clutter
6	Urban scenes	Evaluate performance amid visual background clutter

Table 7.7: Planned experimental evaluation scenarios (reporting template; no results recorded yet)

7.5 Vision-Language Model Results

This section reports the completed zero-shot evaluation of the four VLMs described in Chapter 6, Section 6.6, run on the full IR (10,886 images) and RGB (9,976 images) evaluation sets under the identical protocol described there.

7.5.1 Overall Metrics by Model

Model	Modality	Accuracy	Macro F1	Latency (ms/img)	Invalid Rate
InternVL2-8B	IR	9.56%	9.65%	253.23	0.00%
InternVL2-8B	RGB	66.36%	57.32%*	285.53	14.30%†
DeepSeek-VL-7B-Chat	IR	24.44%	15.98%	257.17	0.00%
DeepSeek-VL-7B-Chat	RGB	72.91%	64.11%	278.18	0.00%
BLIP-2	IR	0.73%	0.29%	264.72	17.45%
BLIP-2	RGB	17.01%	7.27%	N/R	N/R
Qwen2.5-VL-7B	IR	10.89%	12.27%	1136.05	71.90%
Qwen2.5-VL-7B	RGB	78.09%	66.18%	2159.69	9.18%

Table 7.8: VLM overall performance – IR vs. RGB. *Computed as the unweighted mean of the four object-class F1-scores; the RGB set contains no ground-truth *none* images so this class is excluded. †Consists entirely of *none* predictions for the same reason. N/R: not reported in the source evaluation for this model/modality combination.

All four models perform substantially better on RGB than on IR, with accuracy gaps ranging from approximately 16 to 67 percentage points depending on the model – evidence that thermal aerial-object recognition is out-of-distribution for general-purpose

VLM vision encoders (Chapter 8, Section 8.5), consistent with the domain-shift concerns raised for the supervised detectors (Chapter 8, Section 8.3).

7.5.2 IR and RGB Rankings

Rank	Model	Accuracy	Macro F1	Latency (ms)	Invalid Rate
1	DeepSeek-VL-7B-Chat	24.44%	15.98%	257.17	0.00%
2	Qwen2.5-VL-7B	10.89%	12.27%	1136.05	71.90%
3	InternVL2-8B	9.56%	9.65%	253.23	0.00%
4	BLIP-2	0.73%	0.29%	264.72	17.45%

Table 7.9: IR modality – models ranked by accuracy

Rank	Model	Accuracy	Macro F1	Latency (ms)	Invalid Rate
1	Qwen2.5-VL-7B	78.09%	66.18%	2159.69	9.18%
2	DeepSeek-VL-7B-Chat	72.91%	64.11%	278.18	0.00%
3	InternVL2-8B	66.36%	57.32%	285.53	14.30%*
4	BLIP-2	17.01%	7.27%	N/R	N/R

Table 7.10: RGB modality – models ranked by accuracy. *Entirely *none* predictions; see Table 7.8 note.

Qwen2.5-VL-7B is the strongest RGB classifier but incurs 4–8× the latency of the other three models and becomes highly unreliable on IR, where 71.90% of its responses cannot be parsed into any of the five valid classes. DeepSeek-VL-7B-Chat is the only model to record a 0.00% invalid-output rate on both modalities and ranks first on IR and second on RGB, making it the most consistent performer across both sensing modalities.

Rank	Model	IR Accuracy	RGB Accuracy	Mean Accuracy
1	DeepSeek-VL-7B-Chat	24.44%	72.91%	48.68%
2	Qwen2.5-VL-7B	10.89%	78.09%	44.49%
3	InternVL2-8B	9.56%	66.36%	37.96%
4	BLIP-2	0.73%	17.01%	8.87%

Table 7.11: Overall ranking – mean of IR and RGB accuracy

DeepSeek-VL-7B-Chat is identified as the best-performing VLM overall, combining competitive accuracy on both modalities, zero invalid outputs, and among the lowest inference latency of the four models evaluated.

7.5.3 Shared Failure Patterns

Several failure modes recur across models and are worth noting independently of individual results:

- ***None*-collapse on IR.** InternVL2-8B (75.83%), Qwen2.5-VL-7B (71.30%), and BLIP-2 (82.55%) default overwhelmingly to *none* on thermal input; DeepSeek-VL-7B-Chat shows the same tendency but less severely (57.31%), consistent with its higher IR accuracy. This suggests thermal signatures are not recognized as objects at all by vision encoders trained predominantly on RGB imagery, rather than being confused between object categories.
- **Drone recall lags precision on RGB across all models.** For InternVL2-8B, drone precision is 0.904 against recall of only 0.494, with the majority of true-drone images predicted as *none* or misclassified as airplane. Since drone is the majority class in both datasets (≈51%), this mirrors the Drone-vs-background confusion pattern observed for supervised detectors in the pilot study (Chapter 6, Section 6.4.3) and suggests small/distant aerial objects are systematically under-detected regardless of architecture.
- **BLIP-2 single-class collapse.** BLIP-2 predicts *bird* for all 9,976 RGB images and *none*/*drone* almost exclusively on IR (82.55%/17.45%), a qualitatively different failure from confusion-driven errors in the other three models – indicating a breakdown in instruction-grounding rather than fine-grained class confusion. BLIP-2 is not viable as a zero-shot classifier for this task in its current form.
- **Qwen2.5-VL-7B instability under distribution shift.** Its high IR invalid-output rate (71.90%), combined with markedly elevated latency relative to the other three models, suggests instruction-following degrades on out-of-distribution (IR) input even though it is the strongest in-distribution (RGB) performer.

7.6 Comparative Summary

Table 7.12 draws together the current state of both model families evaluated in this project. All three evaluation phases – the pilot study, full-dataset detector training, and zero-shot VLM evaluation – are now complete.

Model Family	Status	Best RGB	Best IR	Notes
Supervised Detectors (pilot)	Complete	YOLOv12x (mAP50 0.849)	YOLOv10x (F1 0.622)	Selected architectures
Supervised Detectors (full dataset)	Complete	YOLOv12x (mAP50 0.9411, mAP50-95 0.6466)	YOLOv10x (F1 0.7627, mAP50-95 0.3381)	YOLOv10x: 59/100 epochs, 4 interrupted runs
VLMs (zero-shot, full dataset)	Complete	Qwen2.5-VL-7B (78.09% acc.)	DeepSeek-VL-7B-Chat (24.44% acc.)	DeepSeek best overall (mean acc. 48.68%)

Table 7.12: Current evaluation status by model family

This comparison reinforces the motivation set out in Chapter 4: closed-vocabulary detectors substantially outperform zero-shot VLMs in raw accuracy on in-distribution imagery – YOLOv12x's 93.29% RGB precision and 94.11% mAP50 dwarf even the strongest VLM's 78.09% RGB accuracy – but VLMs offer a class-agnostic fallback whose practical reliability (invalid-output rate, latency) varies sharply by model and modality. With full-dataset results now available for both detector families and the Adaptive Fallback pipeline implemented in SkyEye (Chapter 6, Section 6.7), the confidence-gated fallback design (Chapter 4, Section 4.8) can be revisited using real detector confidence distributions rather than a hand-set default threshold.

8. Threats to Validity

This chapter discusses factors that may affect the validity, generalizability, and interpretation of the experimental results, arising primarily from dataset composition, modality differences, model behaviour, evaluation methodology, and computational constraints.

Threat	**Summary**
Dataset Bias	Public sources may not represent the full diversity of real-world airborne targets and environments.
Class Imbalance	Uneven sample counts and visual diversity across classes may influence model performance.
Domain Shift	Performance may decline when deployment imagery differs from training data in environment, sensor, scale, or object type.
Thermal–RGB Differences	Differing visual characteristics make direct cross-modality comparison non-trivial.
VLM Hallucination	VLMs may generate incorrect, ambiguous, or unsupported predictions.
Invalid VLM Outputs	Generated responses may not follow the required class vocabulary and must be handled separately.
Computational Cost	Large detectors and VLMs introduce substantial memory and latency overhead.
Confidence Calibration	Detector and VLM confidence scores are not directly comparable, affecting confidence-based selection.

Table 8.1: Summary of threats to validity

8.1 Dataset Bias

Although combining multiple public sources increases visual diversity, the resulting distribution may still differ from real-world surveillance conditions. Camera viewpoint, object scale, background, weather, altitude, and geographic location may be unevenly represented, so strong test-set performance does not guarantee equivalent operational performance.

8.2 Class Distribution

The dataset exhibits a natural class imbalance, with the drone class representing 51.2% of the total images, compared with 18.6% for bird, 17.5% for airplane, and 12.7% for helicopter. This distribution reflects the composition and availability of the source datasets.

The class imbalance may influence model learning and aggregate evaluation metrics, particularly for the less represented classes. To account for this effect, model performance is evaluated using per-class precision, recall, F1-score, and Jaccard scores in addition to micro- and macro-averaged metrics. Macro-averaged metrics are particularly useful because they assign equal importance to each target class regardless of its frequency.

8.3 Domain Shift

Differences between the experimental dataset and unseen deployment environments — illumination, weather, clutter, camera characteristics, object distance, and airframe appearance — pose a key threat to generalization. This is especially relevant for the drone class, given the wide variation in size, shape, colour, and configuration across commercial and custom-built UAVs; performance on known examples should not be taken as evidence of generalization to unseen designs.

8.4 RGB–Thermal Modality Differences

RGB and thermal imagery represent different physical properties — colour and texture versus emitted infrared radiation — so cross-modality performance differences cannot be attributed to detector architecture alone; image quality, resolution, annotation availability, and dataset composition also contribute. As discussed in Chapter 7, Section 7.3.5, the lower full-dataset performance observed for the IR detector (YOLOv10x) relative to the RGB detector (YOLOv12x) is attributed principally to the comparatively limited size and diversity of the available IR source data (Chapter 5, Section 5.8), rather than to architectural differences between the two models.

8.5 Vision-Language Model Hallucination

Zero-shot VLMs may produce plausible but incorrect outputs, particularly for small, occluded, ambiguous, or out-of-distribution targets. Generated responses are therefore not treated as ground truth but explicitly compared against known class labels.

8.6 Invalid VLM Outputs

Because VLMs generate natural-language responses, outputs may not map directly to one of the five target classes (e.g. explanations, multiple classes, or unrelated descriptions). Such cases are recorded as invalid predictions rather than forced into an arbitrary class, and the invalid-output rate is reported separately to distinguish classification errors from failures to produce a valid response. This effect is particularly pronounced for Qwen2.5-VL-7B on IR imagery, where 71.90% of responses were unparseable (Chapter 7, Section 7.5).

8.7 Computational Cost

YOLO-based detectors are designed for efficiency, whereas large VLMs may require substantially more GPU memory and inference time. This matters for practical counter-UAV deployment, where low latency is often required, so higher accuracy does not necessarily imply suitability for real-time use. This tradeoff is directly reflected in the SkyEye tool's three pipeline modes (Chapter 6, Section 6.7), which let the detector-only path be used when latency is critical and the VLM-only or Adaptive Fallback paths be used when contextual accuracy is prioritized.

8.8 Confidence Calibration

Detector confidence scores and VLM-generated probabilities may represent different quantities and are not directly comparable, limiting the confidence-gated detector–VLM architecture now implemented as the Adaptive Fallback pipeline mode (Chapter 6, Section 6.7). Before reliable automatic activation, the threshold should be evaluated on validation data using calibration measures such as the Brier score and Expected Calibration Error (ECE); this calibration step is therefore treated as future work (Chapter 9, Section 9.2, item F3) rather than a universally valid threshold, and the tool currently exposes the cutoff as a user-configurable, hand-set value.

8.9 Evaluation Protocol Limitations

Supervised detectors and zero-shot VLMs solve fundamentally different tasks — localized bounding boxes with confidences versus image-level classification from generated text — so direct numerical comparison between their metrics must be interpreted carefully. The comparison aims to characterize complementary capabilities, not to claim interchangeability.

8.10 Generalizability of Results

Results should be interpreted within the scope of the constructed dataset, target classes, tested models, and experimental hardware. The evaluation of YOLOv8, YOLOv10, YOLO11, YOLO12, RF-DETR, InternVL2, BLIP-2, Qwen2.5-VL, and DeepSeek-VL provides a meaningful comparative benchmark but does not establish universal superiority of any architecture. Further testing on independent datasets, unseen UAV types, additional environmental conditions, and real-time sensor streams – including the operational scenarios in Chapter 7, Table 7.7, none of which have yet been executed – is required before drawing conclusions about operational deployment.

9. Conclusion and Future Work

9.1 Conclusion

This project investigated a multimodal deep-learning framework for detecting and classifying low-altitude aerial objects, with particular emphasis on drones and visually similar airborne targets, motivated by the limitations of closed-vocabulary detectors under varying appearance, environment, and sensing modality.

A standardized dataset was constructed from multiple public sources, converted to a common annotation format around four target classes: airplane, bird, drone, and helicopter, with RGB and thermal imagery retained as separate modalities for modality-specific evaluation.

Five supervised detectors — YOLOv8, YOLOv10, YOLO11, YOLO12, and RF-DETR — were compared in a pilot study to select the strongest architecture per modality, identifying YOLOv12x for RGB and YOLOv10x for IR (Chapter 6, Section 6.4). Both were then trained on the full dataset, reaching 94.11% mAP50 (RGB) and 66.57% mAP50 (IR) respectively (Chapter 7, Section 7.3). In parallel, four VLMs — InternVL2, BLIP-2, Qwen2.5-VL, and DeepSeek-VL — were evaluated zero-shot to assess whether general-purpose multimodal models can classify aerial objects without task-specific training, with prediction distributions and invalid outputs analysed alongside standard classification metrics (Chapter 7, Section 7.5).

Together, these two perspectives — supervised detection for accurate closed-vocabulary localization, and zero-shot VLMs for broader visual interpretation — allowed the results in Chapter 7 to identify each approach's strengths and limitations. The project confirms that no single model or modality is optimal across all conditions: architecture, modality, class distribution, and model behaviour must all inform the design of a practical counter-UAV detection system. IR detection lagged RGB throughout the study primarily due to the more limited availability and diversity of IR source data (Chapter 5, Section 5.8; Chapter 7, Section 7.3.5), rather than a limitation of the detector architecture itself.

Beyond the comparative evaluation, this project delivers **SkyEye**, a locally deployable tool integrating both model families – the trained YOLOv12x (RGB) and YOLOv10x (IR) detectors, and all four evaluated VLMs (with InternVL2 currently disabled pending confirmation, Section 6.7.4) – behind a single Streamlit interface with a working Adaptive Fallback pipeline mode, persistent run history, and batch processing. This closes the loop between the offline comparative evaluation in Chapter 7 and a system that can actually be operated on new imagery, satisfying Objective O8 (Chapter 1, Table 1.2) and forming the foundation for the confidence-based detector–VLM integration and operational scenario testing that remain as future work.

9.2 Future Work

ID	Future Direction
F1	Complete and refine thermal detector training; compare comprehensively with RGB results using a larger, more diverse IR dataset.
F2	Investigate confidence calibration to determine an empirically justified VLM fallback threshold.
F3	Calibrate the Adaptive Fallback confidence threshold (τ) against full-dataset detector confidence distributions using formal calibration measures (Brier score, Expected Calibration Error), replacing the current hand-set cutoff.
F4	Implement background-handling strategies (e.g. blank-annotation training) to reduce false positives.
F5	Investigate RGB–thermal decision-level fusion and compare fusion strategies.
F6	Perform collapse analysis: prediction frequency, dominant-class %, entropy, effective predicted classes, per-class recall.
F7	Analyse high-confidence incorrect predictions at thresholds such as 0.80, 0.90, 0.95.
F8	Extend evaluation to nighttime, fog, clutter, small targets, and unseen drone appearances by executing the operational scenario tests (Chapter 7, Table 7.6) using the completed SkyEye tool.
F9	Improve reproducibility via frozen dataset manifests, model/software versions, and complete inference commands.
F10	Resolve and, if warranted, re-enable InternVL2 in the SkyEye UI, stating explicitly why it is currently disabled and confirming whether it is the same InternVL2-8B checkpoint evaluated in Chapter 7, Section 7.5.
F11	Report confusion matrices and precision/recall/F1-confidence curves for the full-dataset YOLOv12x and YOLOv10x detectors, not available at the time of this revision.

Table 9.1: Future work directions

A. Glossary of Terms

Term	Definition
Closed vocabulary	Fixed, predefined set of classes a model is trained to recognize.
Open vocabulary	Ability to identify categories via natural language rather than a fixed training set.
VLM	A model jointly processing visual and language data for understanding, classification, and reasoning.
Fallback layer	Secondary stage triggered when the primary detector's output is uncertain.
Cascade	A two-stage inference pipeline in which a fast model handles the common case and a slower model is invoked only when the fast model's confidence is insufficient.
Escalation	The act of routing a specific detection from the primary detector to a VLM for re-classification when confidence falls below the configured threshold.
Adaptive Fallback	The SkyEye pipeline mode implementing the confidence-gated cascade between YOLO and a VLM.
Blank annotation	Annotation file with no target objects, representing background-only samples.
RGB	Visible-spectrum imagery (red, green, blue channels).
Thermal / LWIR	Long-wave infrared imagery representing emitted thermal radiation.
Closed-vocabulary detector	Supervised detector restricted to its training classes.
Zero-shot inference	Inference without task-specific training on that data.
Micro F1	F1-score from aggregated predictions across all classes.
Macro F1	Mean of per-class F1-scores.
Jaccard Index	Intersection over union of predicted and ground-truth sets.
Invalid-output	Proportion of outputs unmappable to a

B. Hardware and Software Reference

Component	**Reference**
RGB Data	Visible-spectrum aerial imagery for detector/VLM evaluation
Thermal Data	LWIR imagery for modality-specific evaluation
GPU	NVIDIA RTX 4000 Ada Generation
System RAM	32 GB
CUDA	CUDA 12.1
PyTorch	PyTorch 2.5.1+cu121 with CUDA support
Python	Python 3.12
Object Detectors	YOLOv8, YOLOv10, YOLO11, YOLO12, RF-DETR
Vision-Language Models	InternVL2-8B, DeepSeek-VL-7B-Chat, BLIP-2, Qwen2.5-VL-7B
Annotation Format	YOLO-compatible bounding-box format
Evaluation	Python-based metric and analysis scripts
Experiment Environment	Ubuntu 24.04, local single-GPU development and training environment
Web Interface	Streamlit (SkyEye)
Model Serving	FastAPI + Uvicorn
Database	PostgreSQL 16 + SQLAlchemy
Containerization	Docker Compose
Hosted VLM API	OpenRouter (Qwen2.5-VL)
Quantization	`bitsandbytes` 8-bit

Table B.1: Hardware and software resources used in the project

C. Model and Experiment Summary

Model	Category	Experimental Role
YOLOv8	Object Detector	Pilot-study baseline detector
YOLOv10	Object Detector	Pilot comparison; selected as primary IR detector (YOLOv10x)
YOLO11	Object Detector	Pilot comparison; originally assumed primary detector, superseded
YOLO12	Object Detector	Pilot comparison; selected as primary RGB detector (YOLOv12x)
RF-DETR	Object Detector	Transformer-based pilot comparison
InternVL2	VLM	Zero-shot multimodal reasoning; implemented but currently disabled in the SkyEye UI
BLIP-2	VLM	Zero-shot visual understanding; self-hosted in SkyEye
Qwen2.5-VL	VLM	Zero-shot multimodal reasoning; hosted via OpenRouter in SkyEye
DeepSeek-VL	VLM	Zero-shot multimodal reasoning; self-hosted in SkyEye; best-performing VLM overall

Table C.1: Summary of models evaluated in the project

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